

Railflow — AI-Controlled HVAC

Project schedule and role division · 18 – 28 August 2026

Deadline: Thursday 27 August 2026 (delivery and presentation). Friday 28 August is held as wrap-up buffer and is not planned work. All AI and modelling work completes on **Saturday 22 August**, leaving four clear days for hardware integration, rehearsal and slides.

Roles

Alper	Salma
All code and engineering. Physics simulator, passenger and timetable model, dataset generation, feature engineering, model training, controllers, evaluation, serial protocol implementation, dashboard, and the handoff package. Owns every file in the repository.	Research only. No code. Finds the numbers, standards and prior art that the model currently has to assume. Every output is a short written note with sources, handed to Alper. Primary assignment: HVAC latency — see the research brief on page 3.

Timeline at a glance

Phase	Dates	Days	Owner	Outcome
AI and modelling	Tue 18 – Sat 22 Aug	5	Alper (Salma feeding research)	Trained model, controllers, results, handoff package
Hardware integration	Sun 23 – Mon 24 Aug	2	Team, Alper supporting	Model running on local machine, control command over UART
Rehearsal and slides	Tue 25 – Wed 26 Aug	2	Alper and Salma	Full run-through, backup recording, deck finished
Delivery	Thu 27 Aug	1	Alper and Salma	Presentation and live demo
Buffer	Fri 28 Aug	1	—	Wrap-up only. Nothing scheduled.

The 22 August gate. On that date the AI work is frozen and handed over as a downloadable package. Nothing after 22 August requires retraining or changing the model. If integration reveals a problem, the fix is in the interface layer, not the model — this is what keeps the last four days predictable.

Daily schedule

Blue = Alper (code) · Amber = Salma (research)

Tue 18 Aug		Day 1
ALPER — code	SALMA — research	
M3 — Passenger and timetable model Station list, dwell times, boarding and alighting per stop, morning and evening peaks, door-opening infiltration events. Scenario generator producing a full-day line profile.	HVAC latency — start here Highest-value research task in the project. How long does a rail HVAC unit actually take to respond to a command? See the brief on the next page.	
End of day: A realistic full-day Cairo–Alexandria profile with visible rush-hour structure		
Wed 19 Aug		Day 2
ALPER — code	SALMA — research	
M4a — Dataset and features Scenario loop over seasons, occupancy profiles and stochastic setpoint policies (~30k rows). Lagged commands, EWMA, cyclical time, and the lookahead features: time to next station, expected boarding, +30 min weather.	HVAC latency, continued Manufacturer datasheets, technical papers, thesis work. Then: EN 13129, EN 14750 and UIC 553-1 — what exactly do they specify, and is any of it accessible?	
End of day: Training dataset on disk; first latency findings shared with Alper		
Thu 20 Aug		Day 3
ALPER — code	SALMA — research	
M4b + M5a — Model and controllers Train the forecasting model (chronological split, never random). Verify it beats a persistence baseline. Implement the thermostat baseline and the latency-compensated predictive controller.	Equipment and operating context Real rail HVAC unit specifications: cooling capacity, COP, control interface. Egyptian National Railways fleet and the Cairo–Alexandria corridor.	
End of day: Model beating baseline; both controllers running		
Fri 21 Aug		Day 4
ALPER — code	SALMA — research	
M5b + M6 — Results and lag sweep Energy vs comfort comparison on unseen weather and occupancy. Then sweep the actuator lag across 0, 2, 5, 10, 20 min and plot the dependence. This produces the headline number.	Prior art and presentation material What has been published on predictive or anticipatory HVAC control in rail? Collect figures, diagrams and citable references for the deck.	
End of day: The result: energy saved at equal comfort, plus the lag sensitivity plot		
Sat 22 Aug		Day 5
ALPER — code	SALMA — research	
Handoff package — AI COMPLETE Export the model in a portable format. Wrap inference behind a single <code>predict(state) → control</code> command call. Write and implement the UART protocol. Package it so it runs on a clean machine.	Consolidate Turn all research into one reference document: findings, numbers, sources, and explicitly what could not be found.	
End of day: GATE: model frozen and handed over. Research document delivered.		

Sun 23 Aug		Day 6
ALPER — code	SALMA — research	
Handoff support + dashboard Team downloads the package and runs it locally. Alper is on call for setup problems, and starts the live demo dashboard.	Slide structure Draft the narrative and decide which findings earn a slide. 20 minutes is short.	
End of day: Package running on a machine that is not Alper's		
Mon 24 Aug		Day 7
ALPER — code	SALMA — research	
Dashboard finished Baseline against predictive, side by side, running live. Static plots built first as the guaranteed fallback.	Slide content Write the deck against the real results, not placeholders.	
End of day: Working dashboard; UART carrying control commands		
Tue 25 Aug		Day 8
ALPER — code	SALMA — research	
Full integration test Everything end to end: simulator, model, controller, UART, board, dashboard. Fix what breaks.	Q&A preparation Draft answers to the hard questions, especially anything about assumed numbers.	
End of day: One clean end-to-end run		
Wed 26 Aug		Day 9
ALPER — code	SALMA — research	
Rehearsal and freeze Two timed run-throughs. Record a backup video of the demo. Freeze the code — no changes after today.	Rehearsal Present together, time it, cut whatever does not fit.	
End of day: Everything ready. Backup recording exists.		
Thu 27 Aug		DELIVERY
ALPER — code	SALMA — research	
Presentation and live demo Arrive early, test the hardware on site, keep the backup video ready.	Presentation Co-present. Salma covers standards and context.	
End of day: Delivered		
Fri 28 Aug		Buffer
ALPER — code	SALMA — research	
Wrap-up, repository tidy, README final pass. Deliberately empty so that a slip earlier in the week has somewhere to go.	Wrap-up.	
End of day: —		

Research brief for Salma

Written as questions rather than topics, so it is always clear when an item is finished.

The point of this work. The simulator currently has to *assume* several numbers because no public source was found for them. Each assumption is a question the jury could ask that we cannot fully answer. Every number Salma finds converts an assumption into a citation. Where nothing exists, **recording that clearly is also a result** — it is what justifies our fallback method.

Format for every item: the number or finding, the source (link, standard clause, or document title and page), and a one-line note on how much to trust it. A short written note beats a long conversation.

Priority 1 — HVAC latency (18–19 Aug)

This is the assigned primary topic and the single most valuable thing to find. It is the one number our headline result depends on that we currently cannot source.

Question	Why it matters	Where to look
When an HVAC unit receives a command, how long before cooling output actually starts? (dead time — compressor spin-up, damper travel)	Currently assumed at 2 min with no source	Manufacturer datasheets and manuals; HVAC control textbooks; compressor start-up specifications
How long to reach full output once it starts? (time constant — coil and refrigerant inertia)	Currently assumed at 5 min with no source. Drives the entire result.	Rail HVAC manufacturers; academic papers on HVAC dynamics; theses on rail air conditioning
Do EN 13129, EN 14750 or UIC 553-1 specify any response time, pull-down time or settling requirement?	A standards-backed number would be the strongest possible answer	Standards catalogues, university library access, published summaries and course notes
Has anyone published a measured temperature response curve for a real train cabin?	Would let us validate the simulator against reality rather than only against physics	Google Scholar, ResearchGate, ScienceDirect: "train cabin thermal response", "railway HVAC dynamic model", "pull-down test"
What is a typical pull-down time — how long to cool a hot parked coach to comfort?	An indirect but usable check on the whole model	Commissioning documents, UIC 553-1 test descriptions, manufacturer claims

Important — this cannot fail. An earlier search already suggested these numbers may simply not be public. That is fine and it is planned for: the model sweeps the lag across 0, 2, 5, 10 and 20 minutes and reports how the benefit depends on it, rather than claiming a value. So Salma's finding is **upside, never a blocker**. If the answer is "not publicly available", write that down with the searches tried — it becomes the slide that justifies our approach.

Priority 2 — Standards and equipment (19–20 Aug)

Question	Why it matters
EN 13129 — what fresh air rate per passenger is mandated?	We assume 15 m ³ /h/passenger. A real figure upgrades it instantly, with no vehicle data needed.
EN 13129 — what is the interior temperature curve as a function of outdoor temperature?	We use a simplified linear ramp (22 °C to 26 °C). The real curve would replace it.
UIC 553-1 — confirm the climatic chamber test conditions	We claim Egypt exceeds the standard's bounds. This claim needs to be right.
What cooling capacity do real mainline coach HVAC units have?	We use 40 kW, at the top of the commercial range. Worth confirming.
What COP or EER do these units achieve at high ambient temperature?	Our COP curve shape is assumed. Affects every energy number.

Priority 3 — Context and prior art (20–21 Aug)

Question	Why it matters
Egyptian National Railways: fleet on the Cairo–Alexandria corridor, journey time, typical loading	Makes the scenario concrete and credible to a rail audience
Has predictive or timetable-aware HVAC control been published for rail?	We need to know whether we are novel or reinventing. Either answer is useful — and it is far better to know before the jury does.
How much of a train's total energy goes to HVAC?	Turns a percentage saving into a number that means something
Are there published complaints or studies about cooling on Egyptian trains?	Grounds the problem in something real rather than hypothetical

The 22 August handoff package

What Alper delivers so the prototype can be built without him in the loop.

Item	What it is
Trained model file	Portable format, no version-fragile pickles. Loads on any machine.
Inference wrapper	One function: current conditions in, control command out . No knowledge of the model internals needed to use it.
Serial protocol document	Exact UART frame: field order, units, ranges, message rate. Written so the board side can be implemented from the document alone.
Serial sender	Working code, validated against a virtual COM port so it is testable without the board.
Setup instructions	Clone, install, run. Verified on a machine that is not Alper's.

Checkpoints

Date	Checkpoint	If it is missed
Wed 19 Aug	Training dataset generated	Cut scenario variety, not model quality. One season is enough.
Thu 20 Aug	Model beats the persistence baseline	Check the lookahead features are actually wired in — that is almost always the cause.
Fri 21 Aug	Headline result exists	Drop the lag sweep to a single value and restore it later. Protect the main comparison.
Sat 22 Aug	AI complete and handed over	Hard gate. Ship what works. An imperfect model handed over on time beats a better one that arrives after integration has stalled.
Mon 24 Aug	Control command reaching the board	Demo the dashboard alone. The result stands without hardware.
Wed 26 Aug	Rehearsed, frozen, backup recorded	Present from the recording. Never debug live.

Main risks

Risk	Handling
Hardware or board is not ready in time	The dashboard demo is independent of UART and stands on its own. Hardware is an enhancement to the demo, never a dependency.
Live demo fails in the room	A backup recording is made on 26 Aug. This is not optional.
Latency research finds nothing	Already the assumed case. The lag sweep is the designed answer, and it is arguably a stronger position than a single sourced number.
Model quality disappoints	The comparison is against a thermostat baseline on identical scenarios, so the result is relative. It stays meaningful even if absolute accuracy is modest.
Scope creep after 22 Aug	The gate exists precisely to prevent this. Anything new goes on a 'later' list, not into the build.
Both people converge on the same task	Roles do not overlap by design: Alper writes every line of code, Salma writes no code. Nothing needs negotiating mid-week.